

INTRODUCTION

- Pollution in marine ecosystems has **direct and indirect impacts** on marine biota, terrestrial ecosystems, and humans.
 - **Eight million tons of plastic waste** are added to our oceans every year.¹
 - Microplastics have **high bioaccumulation rates, persistence in the environment, and negative impacts on marine wildlife.**²
 - **Impacts of microplastic ingestion** by organisms include laceration, inflammation, damage to the digestive tract, and disruption of reproductive behavior.³
- Images of *Artemia salina* in the wild.
- Figure 1: Adult *Artemia salina*
- Figure 2: Adult *Artemia salina*
- <https://aquariumbreeder.com/brine-shrimp-life-cycle-benefits-div-hatchery/>
- *Artemia salina* (Brine Shrimp; Figures 1 and 2) were chosen as the model organism for this experiment because they are filter feeders, have definitive lifecycle milestones, and are prey (a food source) for many organisms in marine ecosystems.

EXPERIMENT SUMMARY

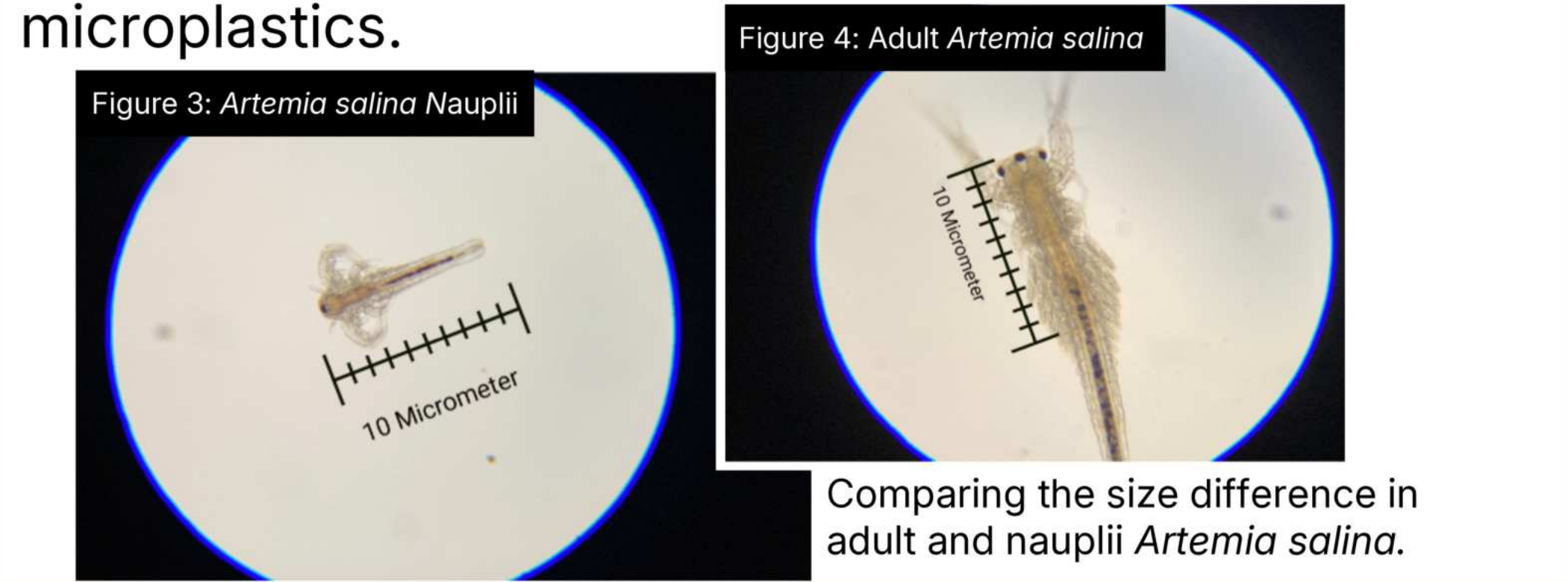
Varying **microplastic solution concentrations** and varying **exposure durations** of microplastic particles to *Artemia salina*, both exposed as nauplii (adolescents; Figure 3) and hatched in microplastic solutions.

Independent Variables (IV):

- The microplastic concentration in artificial seawater (0.2, 0.02, and 0.002 grams / 200 mL).
- The microplastic exposure duration (1, 2, 3, 48, 96, 144, and 192 hours; plus 24 hours for *Artemia salina* hatched in microplastic solutions).
- The points in the lifecycle when *Artemia Salina* are exposed to microplastics (exposed as nauplii or hatched in microplastic solution).

Dependent Variable (DV):
The amount of microplastic particles ingested by *Artemia salina*.

Control:
A batch of *Artemia salina* that is not exposed to microplastics.



HYPOTHESES

The amount of ingested microplastic particles in *Artemia salina* will increase as the concentration of microplastic particles in solution increases and / or as the exposure duration increases.

The amount of ingested microplastics in *Artemia salina* hatched in microplastic solutions (exposed for their entire lifecycle) will be greater than the amount of ingested microplastics in *Artemia salina* exposed to microplastics as nauplii (exposed later in the lifecycle).

DETERMINING THE RELATIONSHIP BETWEEN AMOUNTS OF INGESTED MICROPLASTIC PARTICLES AND MICROPLASTIC SOLUTION CONCENTRATION, EXPOSURE DURATION, AND THE POINT IN THE LIFECYCLE WHEN ARTEMIA SALINA (BRINE SHRIMP) ARE EXPOSED TO MICROPLASTICS

RESEARCH QUESTION

Is there a relationship between the **quantity of microplastic particles ingested** by *Artemia salina* with varied microplastic solution concentrations, varied exposure durations, and different points in the lifecycle when exposed?

METHODOLOGY

INCUBATION AND HATCHING 01

Artemia salina hatched in microplastic solutions were cultured in three different small **brine shrimp hatcheries** that contained different microplastic concentrations: 0.2, 0.02, and 0.002 g / 200 mL (Figure 5). *Artemia salina* exposed to microplastics as nauplii were hatched in a large hatchery with artificial seawater (no microplastics; Figure 6).

Figure 5: Small Brine Shrimp Hatcheries

Figure 6: Large Brine Shrimp Hatcheries

ADDITIONAL IMAGES

Figure 12: Imaged *Artemia salina*

02 MICROPLASTIC EXPOSURE

Artemia salina hatched in the microplastic solutions remained in those solutions and then were transferred to 200 mL containers. Cultured *Artemia salina* hatched in artificial seawater were exposed as nauplii to **0.2g, 0.02 g, and 0.002g / 200 mL** of microplastics in 200 mL containers (Figures 7 and 8).

Figure 7: Different Microplastic Concentrations

Figure 8: *Artemia salina* Nauplii

03 TAKING TIME SAMPLES

At the **1, 2, 3, 48, 96, 144, and 192 hour marks**, three specimens of both *Artemia salina* hatched in microplastic solutions and *Artemia salina* exposed as nauplii were extracted from their cultures with a glass pipette and placed in **well plates** labeled appropriately (Figure 9).

Figure 9: Graphic of *Artemia salina* Well Plate

04 CHEMICAL FREEZING

Both *Artemia salina* hatched in microplastic concentrations and exposed as nauplii were dehydrated following the **Glutaraldehyde Fixation Protocol** and then an **ethanol dilution series** was conducted to dehydrate and "freeze" the *Artemia salina* specimens, so they could be imaged later.

Figure 10: All Chemicals Used for the Glutaraldehyde Fixation Protocol

05 IMAGING AND COMPARING

A **fluorescent microscope** and **ImageJ** software were used to measure the area of fluorescent microplastics inside the dehydrated *Artemia salina* specimens for both *Artemia salina* hatched in microplastic concentrations and exposed as nauplii (Figure 11).

Figure 11: *Artemia salina* with Ingested Microplastics on ImageJ

RESULTS

The imaging process used the area of fluorescence in each image to indicate the presence and amount of microplastic intake. All control samples had no fluorescence indicating zero microplastic intake, which was used as a baseline.

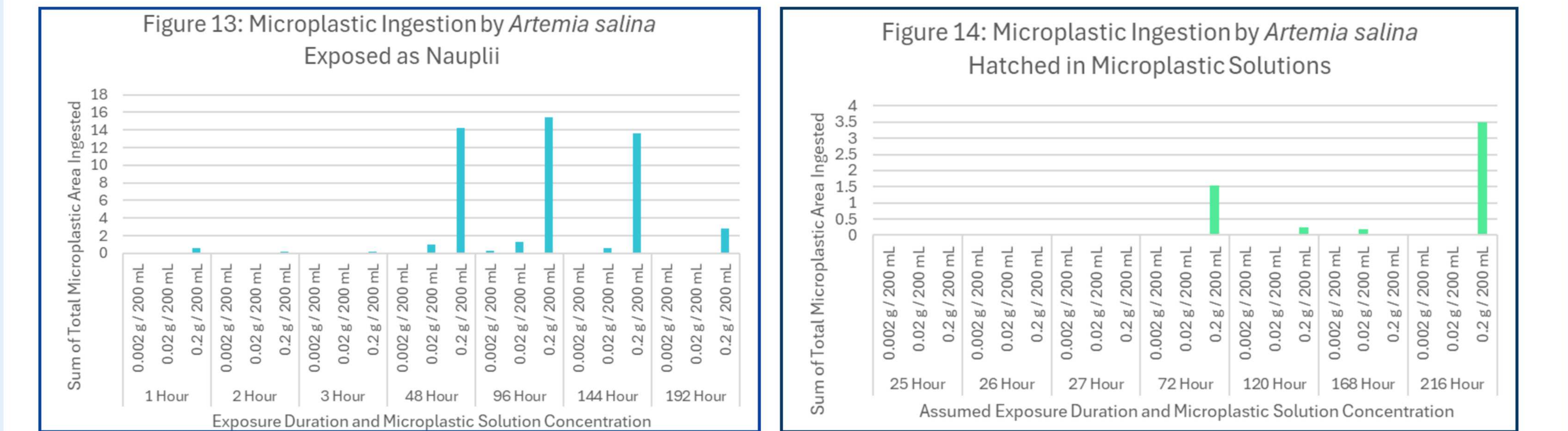


Figure 13: This graph presents the experimental samples of *Artemia salina* exposed to microplastics as nauplii by exposure duration and microplastic concentration.

Figure 14: This graph presents the experimental samples of *Artemia salina* hatched in microplastic solutions by exposure duration (24 hours after assumed hatching) and microplastic concentration.

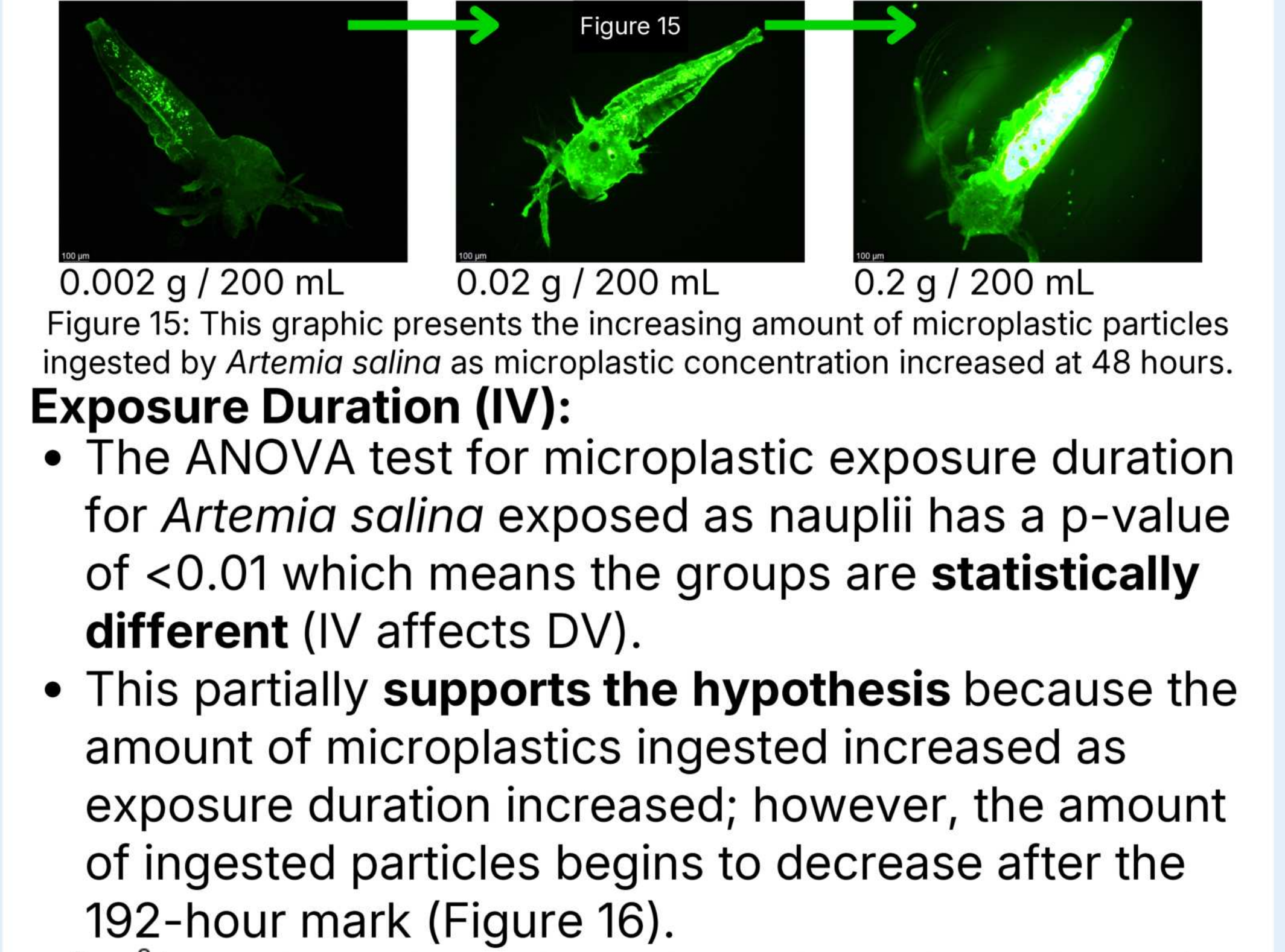
In both populations, the amount of microplastic particles ingested increased as both microplastic concentration increased and exposure duration increased. The amount of ingested microplastics for *Artemia salina* exposed as nauplii was greater than the amount of ingested microplastics for *Artemia salina* hatched in microplastic solutions.

CONCLUSION

There are statistical relationships between microplastic concentration, exposure duration, and point in the lifecycle (Independent Variables) with the amount of microplastic particles ingested (Dependent Variable).

Microplastic Concentration (IV):

- The ANOVA test for microplastic concentrations (0.2, 0.02, and 0.002 grams / 200 mL) has a p-value of <0.01, which means the groups are **statistically different** (IV affects DV).
- This **supports the hypothesis** because there was an increase in the amount of ingested microplastics as the microplastic concentration increased.



Exposure Duration (IV):

- The ANOVA test for microplastic exposure duration for *Artemia salina* exposed as nauplii has a p-value of <0.01 which means the groups are **statistically different** (IV affects DV).
- This partially **supports the hypothesis** because the amount of microplastics ingested increased as exposure duration increased; however, the amount of ingested particles begins to decrease after the 192-hour mark (Figure 16).

Point in Lifecycle (IV):

- The ANOVA test comparing *Artemia salina* hatched in microplastic solutions to those exposed to microplastics as nauplii has a p-value of <0.01 which means the groups are **statistically different** (IV affects DV).
- **The hypothesis is not supported** because it was found that *Artemia salina* hatched in microplastic solutions ingested fewer microplastics than *Artemia salina* exposed as nauplii (Figures 13 and 14).
- **Possible reasoning** for *Artemia salina* exposed as nauplii ingesting more microplastics is that they acted as a **new stimulus** to the *Artemia salina*.

FUTURE RESEARCH

This research will inform measuring the **transference of microplastics** from *Artemia salina*, a primary consumer, to Actinaria (Sea Anemones), a secondary consumer. Future research is planned for 2026-2027.

REFERENCES

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5. The uptake and elimination of polystyrene microplastics by the brine shrimp, *Artemia parthenogenetica*, and its impact on its feeding behavior and intestinal histology. *Chemosphere*.